



**Politecnico
di Torino**

Politecnico di Torino

Master of Science in Energy and Nuclear Engineering

01RARXY - Nuclear Fission Plants

A.Y. 2025/2026

FLOW PATTERNS AND TWO-PHASE PRESSURE DROPS

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1 Introduction

1.1 Laboratory Goals

The present laboratory activity is focused on the investigation of flow patterns and pressure drops in gas–liquid two-phase flow. The experiments are carried out by varying the upward flow rates of air and water, allowing the reproduction of different operating conditions within a vertical plexiglass pipe.

The main objective of the laboratory is to simulate the behavior of an air–water mixture in a controlled environment and to analyze the resulting flow regimes. Particular attention is devoted to the identification of flow pattern types, the measurement of the average void fraction within a section of the pipe, and the evaluation of pressure changes along the vertical test section.

The experimental results are systematically compared with the predictions provided by well-established theoretical models and empirical correlations. Flow pattern observations are analyzed with reference to widely adopted maps, such as those proposed by Hewitt–Roberts and Taitel–Dukler, in order to assess their predictive capability. Furthermore, the void fraction is evaluated using different approaches, including the homogeneous model and correlations such as Zivi, Chisholm, CISE, and the Drift Flux model, and subsequently compared with experimental measurements.

In addition, pressure drops are analyzed by distinguishing between elevation and frictional contributions. The elevation component is estimated using the void fraction obtained from the various models, while the frictional pressure drop is evaluated using both the homogeneous and Friedel models. Different combinations of these models are then employed to estimate the total pressure change, enabling a comprehensive comparison between experimental data and theoretical predictions.

The overall aim of this work is to provide a deeper understanding of two-phase flow behavior in vertical configurations, highlighting the strengths and limitations of the adopted models through a detailed comparison with experimental observations.

1.2 Experimental Facility and Instrumentation

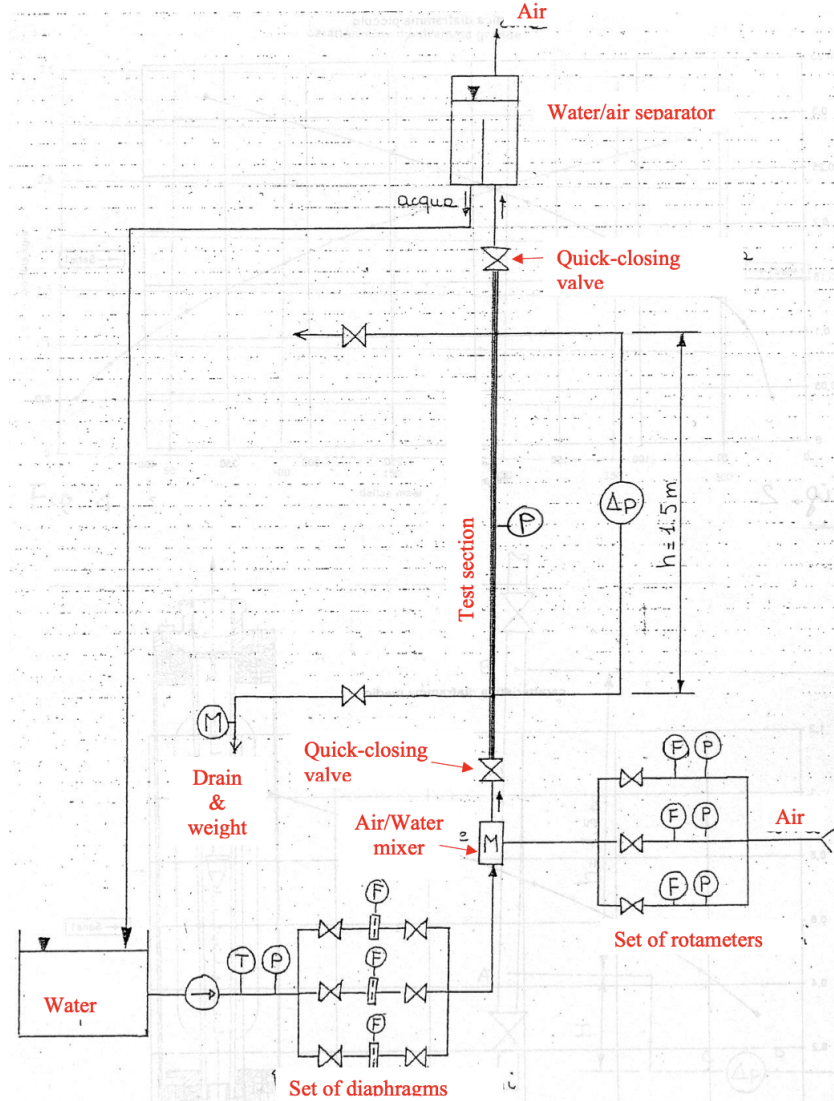


Figure 1: Sketch of the experimental facility

Figure 1 illustrates a schematic representation of the experimental facility operating under upward two-phase flow conditions. Air and water are mixed at the inlet section of the vertical test pipe through a porous bronze distributor (M), which ensures a uniform dispersion of the gas phase within the liquid stream and promotes the formation of a stable air–water mixture.

The air supply is provided by a compressed air system, whereas water is continuously delivered by pumps drawing from a storage tank positioned at the base of the facility. After passing through the test section, the water is discharged and recirculated back into the same tank, thus allowing continuous operation of the experimental loop.

The resulting air–water mixture flows upward through a vertical pipe with an internal diameter of 26 mm. Along the test section, two synchronized fast-closing valves are installed at the inlet and outlet of the vertical pipe. These valves are capable of shutting simultaneously within a very short time interval, thereby trapping a finite volume of the flowing two-phase mixture inside the pipe segment between them. By measuring the mass of water retained within this isolated section, the average void fraction of the two-phase flow can be experimentally determined with good accuracy.

At the outlet of the vertical pipe, the two-phase mixture enters a vertical cylindrical separator. Inside

this component, phase separation occurs due to gravity: the air phase exits from the upper section of the separator, while the liquid phase is collected at the bottom and subsequently recirculated to the storage tank. This arrangement enables stable operating conditions and continuous monitoring of the upward air–water flow system.

The experimental facility is equipped with a dedicated instrumentation system whose purpose is to support the main measurements required by the laboratory: setting the inlet flow rates, determining fluid properties, evaluating the void fraction, and measuring the pressure change along the test section. Following the goal-oriented organization adopted in the reference laboratory report, the instrumentation can be grouped as follows.

- **Temperature measurement and property evaluation.** A PT100 resistance temperature detector (RTD) is installed upstream of the diaphragm assembly used for water flow-rate measurements. The measured water temperature is used to evaluate the liquid thermophysical properties, in particular water viscosity.
- **Water-flow-rate determination.** The liquid flow rate is measured by means of three interchangeable diaphragms mounted on parallel branches, selected according to the required operating range:

$$0.12\text{--}0.32 \text{ kg/s}, \quad 0.30\text{--}1.10 \text{ kg/s}, \quad 0.50\text{--}3.00 \text{ kg/s}.$$

For each diaphragm, the pressure difference is measured by a Rosemount differential pressure transducer with a full-scale range of 0.37 bar. The corresponding calibration curves are then used to convert the measured pressure drop into the water mass flow rate.

- **Air-flow-rate determination.** The gas flow rate is monitored using three rotameters installed on parallel branches, covering the ranges

$$0.2\text{--}1.2 \text{ Nm}^3/\text{h}, \quad 0.45\text{--}4.2 \text{ Nm}^3/\text{h}, \quad 1.5\text{--}17.0 \text{ Nm}^3/\text{h}.$$

Since the rotameter readings are expressed in normal cubic meters per hour, referring to air at 20°C and 1.013 bar, they are corrected using the local inlet pressure and temperature conditions to obtain the actual air mass flow rate.

- **Operating-pressure monitoring.** Bourdon-type relative pressure gauges are used to check the operating conditions of both supply lines. Gauges with a full-scale range of 6 barg are positioned downstream of the pump and upstream of the water diaphragm section, while gauges with a full-scale range of 8 barg are installed upstream of the rotameters to measure the air supply pressure for the calculation of the actual air mass flow rate.
- **Pressure-drop measurement along the test section.** A Rosemount differential pressure transducer with a full-scale range of 0.37 bar measures the pressure variation between the taps located along the vertical pipe. The distances between the taps, reported in Figure 1, are required to convert the measured pressure difference into the corresponding pressure gradient.
- **Reference pressure measurement.** A Rosemount gauge pressure transducer with a full-scale range of 1.5 bar measures the pressure at approximately the mid-height of the test section. This value is used to define the local operating pressure of the air–water mixture needed to evaluate the gas density under the actual experimental conditions.
- **Void-fraction measurement.** Two synchronized fast-closing valves isolate a known volume of the vertical pipe and trap the flowing two-phase mixture. By weighing the water retained inside this section after valve closure, the average void fraction can be determined as shown below. [1]

2 Measured Experimental Data Evaluation

2.1 Fluid and Geometrical Properties

Before evaluating the measured flow rates and the derived two-phase quantities, the main fluid properties and geometrical parameters of the test section are defined. The reference values adopted in the calculations are summarized in Table 1.

Quantity	Value	Unit
Liquid density, ρ_l	1000	kg/m ³
Air specific gas constant, R_{air}	287	Pa m ³ /(kg K)
Air temperature, T_{air}	293.15	K
Pressure-tap distance, h	1.5	m
Pipe internal diameter, D	0.026	m
Pipe cross-sectional area, A	5.309×10^{-4}	m ²
Surface tension, σ	0.072	N/m
Air dynamic viscosity, μ_g	1.81×10^{-5}	Pa s

Table 1: Fluid and geometrical properties used in the data reduction

The air temperature is assumed to remain equal to 20°C during the experiments, corresponding to $T_{air} = 293.15$ K. Under this assumption, the air density is evaluated from the ideal gas law using the absolute pressure measured in the test section:

$$\rho_g = \frac{p_{abs}}{R_{air}T_{air}} \quad (1)$$

where p_{abs} is obtained from the measured gauge pressure by adding the atmospheric pressure:

$$p_{abs} = (p_{rel} + 1.013) 10^5 \quad [\text{Pa}] \quad (2)$$

The dynamic viscosity of water is evaluated as a function of the measured water temperature T_l , expressed in °C, through the following empirical correlation:

$$\mu_l = 0.001002 10^{\frac{1.3272(20-T_l) - 0.001053(T_l-20)^2}{T_l+105}} \quad [\text{Pa s}] \quad (3)$$

The air dynamic viscosity is instead assumed constant at $\mu_g = 1.81 \times 10^{-5}$ Pa s, consistently with the reference air temperature of 20°C.

2.2 Water Flow Rate $\dot{W}$

The water flow rate is measured by diaphragms. The diaphragms used in the experimental setup are equipped with corner pressure taps, and their geometric characteristics fall outside the applicability range of the correlations provided by the CNR-UNI and ISO standards. Therefore, each diaphragm was individually calibrated in the laboratory by measuring the pressure drop across the diaphragm and determining the corresponding mass flow rate through the “weight method”. In this procedure, the flow rate is obtained by measuring the mass of water collected over a known time interval.

As a result of the calibration procedure, the water mass flow rate can be evaluated through the following relationship:

$$\dot{W} = \alpha_{MQ} \frac{\pi d^2}{4} \sqrt{2\rho\Delta p} + B \quad (\text{kg/s}) \quad (4)$$

where Δp is the differential pressure across the diaphragm, expressed in Pascal.

The calibration constants α_{MQ} and B were determined by means of a best-fit procedure using the least-squares method applied to the experimental calibration data. The resulting values are reported in Table 2.

Diaphragm	D [m]	d [m]	$\beta = d/D$	α_{MQ}	B
Small	0.0168	0.0084	0.5000	0.650729	0.01415274
Medium	0.0212	0.0153	0.7217	0.728193	0.00375718
Large	0.0340	0.0255	0.7500	0.773353	0.00986774

Table 2: Calibration constants for the diaphragms

2.3 Air Flow Rate $\dot{W}_e$

The air flow rate is measured by rotameters. The rotameter consists of a vertical tapered tube, whose cross-sectional area increases in the upward direction, through which the fluid flows upward.

Due to the combined effects of pressure and viscous drag, the fluid exerts an upward force on a float (bullet) placed inside the tube. The float reaches an equilibrium position when its weight is exactly balanced by the drag and friction forces exerted by the flowing fluid.

As the flow rate increases, the float moves upward along the tube, allowing the flow rate to be determined from its position.

The rotameters used in the experimental setup were calibrated with air at a temperature of 20°C and at atmospheric pressure. Consequently, the flow rate values indicated by the instruments, expressed in Nm³/h, refer to these standard operating conditions.

Assuming that the air temperature during the experiments remains approximately equal to 20°C, the actual air mass flow rate can be evaluated through the following relationship:

$$\dot{W}_e = 1.204 \sqrt{\frac{p_{e,\text{rel}} + 1.013}{1.013}} Q_s \quad (5)$$

where:

- $p_{e,\text{rel}}$ is the relative pressure measured by the manometer located at the rotameter inlet, expressed in bar;
- Q_s is the volumetric flow rate indicated by the rotameter under normal conditions, expressed in Nm³/h;
- $\dot{W}_e$ is the actual air mass flow rate.

2.4 Void Fraction α

The average void fraction in the test section can be evaluated by measuring the amount of water trapped inside the pipe when the quick-closing valves are simultaneously shut.

Under these conditions, the average void fraction can be expressed as:

$$1 - \alpha = \frac{V_l}{V_{\text{tot}}} \quad (6)$$

where:

$$V_l = \frac{M_l}{\rho_l} \quad (7)$$

and

$$V_{\text{tot}} = \frac{M_{l0}}{\rho_l} \quad (8)$$

Therefore, the void fraction can be written as:

$$\alpha = 1 - \frac{M_l}{M_{l0}} \quad (9)$$

where:

- M_l is the mass of liquid trapped between the quick-closing valves during two-phase flow conditions;
- M_{l0} is the mass of liquid trapped between the quick-closing valves when no air flows through the test section, i.e. when the test section is completely filled with water;
- ρ_l is the liquid density.

The mass trapped between the quick-closing valves is given by the sum of the mass drained from the test section, M_{dr} , and the residual mass of water remaining inside the test section between the lower valve and the discharge tap, M_{res} :

$$M_l = M_{\text{dr}} + M_{\text{res}} \quad (10)$$

The residual mass was estimated by considering the internal volume of the pipe section that remains filled with water after the discharge procedure and assuming a water density equal to:

$$\rho_l = 1000 \text{ kg/m}^3 \quad (11)$$

The resulting residual mass is approximately:

$$M_{\text{res}} \approx 0.116 \text{ kg} \quad (12)$$

When the test section is completely filled with water (i.e. in the absence of air flow), the mass of water that can be discharged from the test section is:

$$M_{\text{dr},0} = 0.898 \text{ kg} \quad (13)$$

Therefore, the total mass of water trapped between the quick-closing valves under single-phase liquid conditions is:

$$M_{l0} = M_{\text{dr},0} + M_{\text{res}} \quad (14)$$

which yields:

$$M_{l0} = 0.898 + 0.116 = 1.014 \text{ kg} \quad (15)$$

During the experiments performed under two-phase flow conditions, the mass of liquid trapped between the quick-closing valves is evaluated as:

$$M_l = M_{\text{dr}} + M_{\text{res}} \quad (16)$$

2.5 Experimental Data

Test #	Water diaphragm	Air flow rate (Nm ³ /h)	Rotameter air pressure (bar)	Water diaphragm press. drop (mbar)	Relative pressure (bar)	Type of Δp	Pressure drop (mbar)	Water temp. (°C)	Water mass (g)	Flow pattern
1	SMALL	0.25	6.0	363.40	0.31	pC - pD	-	23.4	518	BUBBLY / SLUG
2	SMALL	0.23	6.0	360.01	0.31	pD - pC	44.76	22.77	524	BUBBLY / SLUG
3	SMALL	0.73	5.8	361.53	0.29	pD - pC	64.93	21.1	363	SLUG
4	SMALL	3.40	5.2	360.90	0.29	pD - pC	72.69	19.74	234	SLUG
5	SMALL	15.96	4.4	356.78	0.30	pD - pC	55.61	18.47	117	SLUG / CHURN
6	SMALL	16.08	4.4	66.44	0.27	pD - pC	96.31	19.25	43	ANNULAR
7	SMALL	3.40	5.2	66.41	0.27	pD - pC	92.86	19.64	163	SLUG
8	SMALL	0.79	6.0	66.23	0.27	pD - pC	90.13	19.64	298	SLUG
9	MEDIUM	0.26	6.0	81.03	0.32	pD - pC	26.79	16.81	585	BUBBLY
10	MEDIUM	0.73	6.0	81.31	0.31	pD - pC	37.80	16.71	470	SLUG
11	MEDIUM	3.42	5.0	81.12	0.31	pD - pC	51.44	16.52	291	SLUG / CHURN
12	MEDIUM	8.48	5.4	80.39	0.33	pD - pC	30.17	16.22	239	SLUG
13	MEDIUM	16.05	4.4	79.18	0.36	pD - pC	-	16.22	170	ANNULAR
14	MEDIUM	0.26	6.0	216.74	0.27	pD - pC	3.21	17.01	642	BUBBLY
15	MEDIUM	0.78	6.0	216.36	0.30	pD - pC	9.52	16.61	554	BUBBLY
16	MEDIUM	3.33	5.4	212.92	0.35	pD - pC	10.63	16.52	343	BUBBLY / SLUG
17	MEDIUM	8.49	5.2	208.41	0.39	pC - pD	21.60	16.71	268	SLUG / CHURN
18	MEDIUM	16.22	4.4	203.00	0.45	pC - pD	63.92	16.81	177	ANNULAR (almost)

Table 3: Experimental data

Table 3 summarizes the experimental data collected during a total of 18 tests performed under different operating conditions. For each experimental run, the main hydraulic and thermodynamic parameters of the air-water two-phase flow were recorded, including the selected water diaphragm, the air flow rate, the air supply pressure upstream of the rotameters, the differential pressure across the water diaphragm, the relative pressure within the test section, the measured pressure-drop configuration, the pressure variation along the test section, the water temperature, and the mass of water trapped between the fast-closing valves.

In addition, the observed flow pattern corresponding to each operating condition was identified and reported. The dataset includes several flow regimes, such as bubbly, slug, churn, and annular flow, as well as intermediate transition conditions between different regimes. The collected measurements constitute the experimental basis for the subsequent analysis and interpretation of the two-phase flow behavior within the vertical test section.

3 Comparison of Observed Flow Patterns with Predictions of the Maps by Hewitt-Roberts and Taitel-Dukler

The prediction and classification of flow patterns in gas-liquid two-phase flows represent a fundamental aspect in the analysis of multiphase transport phenomena. Different flow regimes are associated with significantly different hydrodynamic behaviors, pressure losses, phase distributions, and heat-transfer characteristics. For this reason, several experimental and theoretical studies have been carried out over the years in order to identify the transition criteria between the various flow configurations.

The characteristic parameters associated with the different flow patterns have been derived either through purely empirical approaches or through theoretical models supported and validated by experimental observations. In most cases, the available predictive methods combine analytical considerations with empirical correlations obtained from extensive experimental databases.

To facilitate the identification of the flow regimes, graphical flow-pattern maps have been developed. These maps define the regions of existence of the various flow configurations and indicate the transition boundaries between one regime and another. Depending on the complexity of the adopted methodology, flow-pattern maps can generally be classified into two main categories:

- **Simple flow maps**, in which the transition between flow regimes is represented through a single pair of parameters. Typical examples are the Hewitt-Roberts map and the Baker map, where the flow-pattern boundaries are obtained directly from experimental observations and empirical correlations.
- **Complex flow maps**, in which different combinations of parameters are used to describe the transition criteria between regimes. These approaches generally consist of multiple interconnected maps or transition equations, each associated with specific physical mechanisms governing the flow evolution. A representative example is the Taitel-Dukler model, which is based on mechanistic considerations and incorporates several transition criteria for the prediction of two-phase flow regimes [2].

In the following sections, the experimental results obtained during the present campaign are compared with the predictions provided by the Hewitt-Roberts and Taitel-Dukler flow-pattern maps, in order to evaluate their capability in representing the observed flow configurations within the investigated operating conditions.

3.1 Derived Fluid-Dynamic Parameters

Before locating the experimental points on the flow-pattern maps, the measured mass flow rates were converted into the fluid-dynamic quantities required by the map coordinates and following experimental/theoretical models comparisons. The gas mass quality, namely the fraction of the total mass flow rate carried by the gas phase, is

$$x = \frac{\dot{W}_g}{\dot{W}_l + \dot{W}_g}. \quad (17)$$

The phase mass velocities are evaluated directly from the phase mass flow rates and the pipe flow area, and the corresponding superficial velocities are

$$G_l = \frac{\dot{W}_l}{A}, \quad G_g = \frac{\dot{W}_g}{A}, \quad j_l = \frac{G_l}{\rho_l} = \frac{\dot{W}_l}{\rho_l A}, \quad j_g = \frac{G_g}{\rho_g} = \frac{\dot{W}_g}{\rho_g A}. \quad (18)$$

The total superficial velocity and the volumetric quality, used in the Taitel-Dukler maps, are

$$j = j_l + j_g, \quad \beta = \frac{j_g}{j_l + j_g}. \quad (19)$$

When the Martinelli parameter is required, it is evaluated for turbulent flow of both phases in a smooth pipe as Lockhart–Martinelli parameter:

$$\frac{1}{X_1} = \left(\frac{x}{1-x} \right)^{0.9} \left(\frac{\rho_l}{\rho_g} \right)^{0.5} \left(\frac{\mu_g}{\mu_l} \right)^{0.1}. \quad (20)$$

3.2 Hewitt-Roberts Map

For the Hewitt-Roberts map, the experimental points are located using the momentum-flux coordinates

$$X_{HR} = \rho_l j_l^2 = \frac{G_l^2}{\rho_l} \quad (21)$$

and

$$Y_{HR} = \rho_g j_g^2 = \frac{G_g^2}{\rho_g} \quad (22)$$

These coordinates were used to locate the experimental operating points on the Hewitt-Roberts flow pattern map. This procedure allowed the measured conditions to be compared with the flow pattern estimations obtained through direct visual observation in a transparent section of the test pipe.

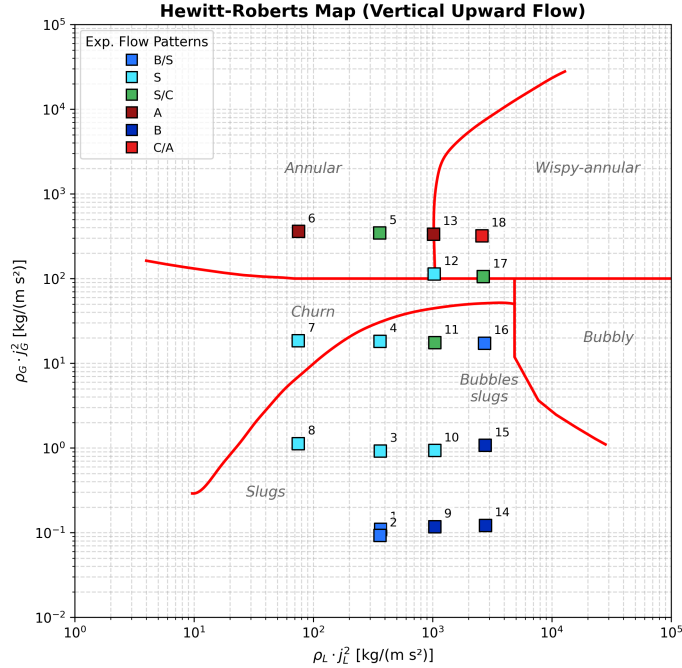


Figure 2: Hewitt-Roberts map with flow pattern predictions

By comparing the results reported in Fig. 2 with the flow patterns visually identified during the experimental campaign and summarized in Table 3, several discrepancies can be observed.

Although the bubbly flow regime was identified three times during the experimental session, it only appears in the outcomes of the implemented model as transition between bubbly and slug. This inconsistency may be attributed to the limited visual distinction between these two flow regimes, which can make their discrimination particularly ambiguous.

Nevertheless, the plug flow regime remains the most frequently observed pattern in both approaches, showing good qualitative agreement.

Overall, these results suggest that visual identification of flow patterns, even if assisted by the analysis of the oscillations of pressure change, is inherently subjective and may not always allow for a fully unambiguous and consistent classification.

3.3 Taitel-Dukler Map

The same experimental operating conditions were also analyzed using the Taitel-Dukler approach. Unlike the Hewitt-Roberts map, which represents the transition between regimes through a single graphical comparison, the Taitel-Dukler model is based on separate mechanistic criteria associated with the physical mechanisms governing the onset of bubbly, slug, churn, and annular flow. For this reason, the comparison is reported through three different maps, each one corresponding to a specific transition region.

For the bubbly-slug/churn transition, the coordinates are one form of dimensionless gas superficial velocity and gas-to-liquid superficial velocity ratio:

$$X_B = j_g \rho_l^{1/2} [g(\rho_l - \rho_g)\sigma]^{-1/4}, \quad Y_B = \frac{j_l}{j_g}. \quad (23)$$

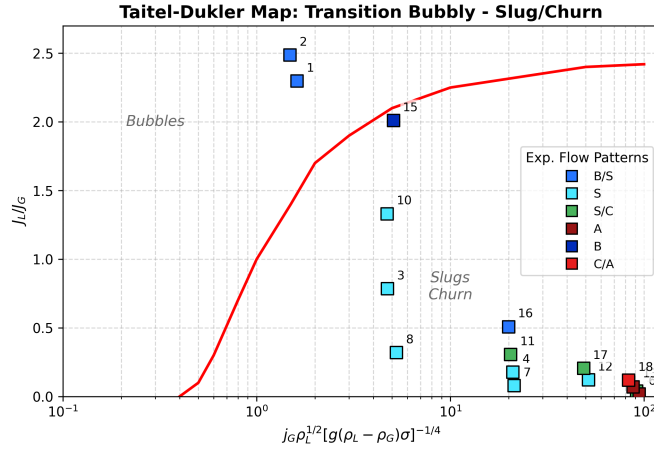


Figure 3: Taitel-Dukler map for the bubbly-flow transition

For the slug-to-churn transition, the coordinates are represented by the volumetric quality and the Froude number:

$$Y_{SC} = \beta = \frac{j_g}{j_l + j_g}, \quad X_{SC} = \frac{j}{\sqrt{gD}}. \quad (24)$$

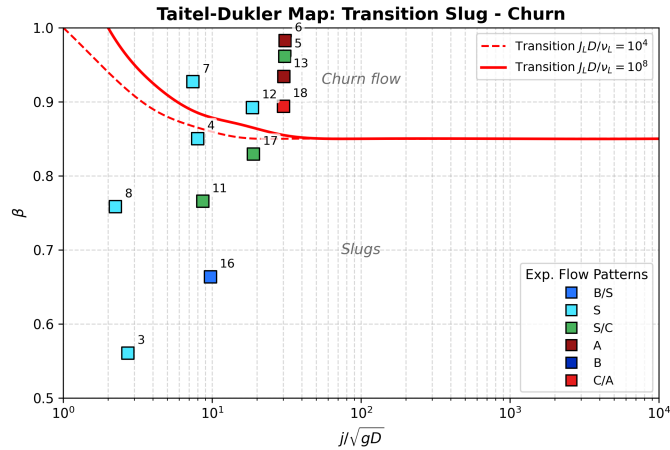


Figure 4: Taitel-Dukler map for the slug-to-churn transition

For the annular-to-slug/churn transition, the horizontal coordinate is the Martinelli parameter calculated as defined above, while the vertical coordinate is the another dimensionless gas superficial velocity:

$$X_A = X_1 = \left[\frac{(dp/dz)_{l,F}}{(dp/dz)_{g,F}} \right]^{1/2}, \quad Y_A = j_g \rho_g^{1/2} [g(\rho_l - \rho_g)\sigma]^{-1/4}. \quad (25)$$

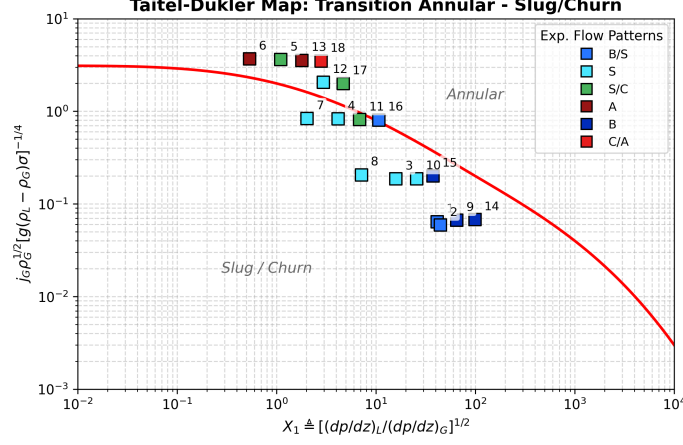


Figure 5: Taitel-Dukler map for the annular-flow transition

Figures 3–5 show that the Taitel-Dukler formulation provides a more detailed interpretation of the experimental observations, since each transition is evaluated through a criterion directly related to the expected flow structure. The bubbly-flow map correctly highlights the operating conditions characterized by low gas flow rates and comparatively high liquid flow rates, while the slug-to-churn transition map identifies the progressive shift toward more intermittent and unstable configurations as the gas superficial velocity increases. Finally, the annular-flow map captures the high-gas-flow-rate conditions for which the gas phase becomes dominant and the liquid phase is mainly distributed near the pipe wall.

The comparison with the visual classifications reported in Table 3 confirms the qualitative trends already observed with the Hewitt-Roberts map, but with a clearer separation between the transition mechanisms. In particular, the intermediate cases classified experimentally as bubbly/slug or slug/churn are located close to the corresponding transition boundaries, which supports the interpretation that these operating points belong to regions where regime identification is intrinsically uncertain. Therefore, the Taitel-Dukler approach complements the Hewitt-Roberts map by providing a more physically based explanation of the observed discrepancies between visual inspection and map prediction.

4 Comparison of Experimental and Theoretical Void Fractions

By definition, the void fraction can be evaluated through the following relationship:

$$\alpha = \frac{1}{1 + \left(\frac{1-x}{x} \right) \left(\frac{\rho_g}{\rho_l} \right) S} \quad (26)$$

where S is the slip ratio, defined as the ratio between the gas velocity and the liquid velocity.

The slip ratio can be evaluated through empirical correlations or theoretical models, generally supported by experimental data. Several models are available in the literature for the prediction of the void fraction under different assumptions regarding the slip ratio. Among the most widely adopted approaches are the following.

- **Homogeneous model.** In the homogeneous approach, the liquid and gas phases are assumed to move with the same velocity. Consequently, the slip ratio is equal to unity:

$$S = 1 \quad (27)$$

and the void fraction becomes:

$$\alpha = \frac{1}{1 + \left(\frac{1-x}{x}\right) \left(\frac{\rho_g}{\rho_l}\right)} \quad (28)$$

- **Zivi correlation.** The Zivi model accounts for the relative motion between the phases by defining the slip ratio as:

$$S = \left(\frac{\rho_l}{\rho_g}\right)^{1/3} \quad (29)$$

Substituting this expression into Eq. (26) allows the corresponding void fraction to be determined.

- **Chisholm correlation.** The Chisholm model, generally considered accurate only at high void fractions, expresses the slip ratio as:

$$S = \sqrt{1 - x \left(1 - \frac{\rho_g}{\rho_l}\right)} \quad (30)$$

- **CISE correlation.** The CISE model provides greater predictive accuracy over a wide range of operating conditions. In this case, the slip ratio is evaluated through:

$$S = 1 + E_1 \left(\frac{y}{1 + yE_2} - yE_2\right)^{0.5} \quad (31)$$

where:

$$y = \frac{\beta}{1 - \beta} \quad (32)$$

$$\beta = \frac{\rho_l x}{\rho_l x + \rho_g (1 - x)} \quad (33)$$

The coefficients E_1 and E_2 are given by:

$$E_1 = 1.578 Re^{-0.19} \left(\frac{\rho_l}{\rho_g}\right)^{0.22} \quad (34)$$

$$E_2 = 0.0273 We Re^{-0.51} \left(\frac{\rho_l}{\rho_g}\right)^{-0.08} \quad (35)$$

The Reynolds and Weber numbers are respectively defined as:

$$Re = \frac{GD}{\mu_l} \quad (36)$$

$$We = \frac{G^2 D}{\sigma \rho_l} \quad (37)$$

- **Drift Flux model.** In the Drift Flux formulation, the void fraction is expressed as:

$$\alpha = \frac{j_g}{C_0 j + u_{gj}} \quad (38)$$

where j_g is the superficial gas velocity, j is the total superficial velocity, C_0 is the distribution parameter, and u_{gj} is the drift velocity.

For bubbly flow:

$$C_0 = 1.13 \quad (39)$$

$$u_{gj} = 1.41 \left[\frac{\sigma g(\rho_l - \rho_g)}{\rho_l^2} \right]^{0.25} \quad (40)$$

For plug flow:

$$C_0 = 1.2 \quad (41)$$

$$u_{gj} = 0.35 \sqrt{\frac{(\rho_l - \rho_g)gD}{\rho_l}} \quad (42)$$

For water at high pressure, the bubbly-flow correlation can generally be applied to all flow regimes with satisfactory accuracy.

Table 4 reports the values of the void fraction, α , predicted by each of the previously described models and compares them with the corresponding experimental values obtained through the procedure discussed in Section 1.3.

For each correlation, the percentage error with respect to the experimental measurement is also reported in order to quantitatively evaluate the predictive accuracy of the different models.

Test #	Experimental	Homogeneous		Zivi		Chisholm		CISE		Drift flux (bubbly)		Drift flux (plug)	
		value	error	value	error	value	error	value	error	value	error	value	error
1	0.375	0.303	19%	0.048	87%	0.266	29%	0.207	45%	0.217	42%	0.216	42%
2	0.369	0.287	22%	0.045	88%	0.254	31%	0.196	47%	0.205	45%	0.204	45%
3	0.528	0.560	6%	0.128	76%	0.458	13%	0.372	30%	0.432	18%	0.422	20%
4	0.655	0.850	30%	0.396	39%	0.688	5%	0.628	4%	0.716	9%	0.683	4%
5	0.770	0.961	25%	0.742	4%	0.833	8%	0.823	7%	0.840	9%	0.793	3%
6	0.843	0.982	17%	0.866	3%	0.885	5%	0.863	2%	0.858	2%	0.811	4%
7	0.725	0.927	28%	0.594	18%	0.776	7%	0.710	2%	0.778	7%	0.743	3%
8	0.592	0.758	28%	0.265	55%	0.607	3%	0.500	16%	0.569	4%	0.559	6%
9	0.309	0.210	32%	0.030	90%	0.191	38%	0.152	51%	0.161	48%	0.157	49%
10	0.422	0.429	2%	0.080	81%	0.362	14%	0.294	30%	0.341	19%	0.330	22%
11	0.599	0.765	28%	0.275	54%	0.613	2%	0.556	7%	0.647	8%	0.617	3%
12	0.650	0.892	37%	0.491	24%	0.732	13%	0.712	9%	0.773	19%	0.732	13%
13	0.718	0.934	30%	0.625	13%	0.786	10%	0.788	10%	0.816	14%	0.771	7%
14	0.252	0.145	43%	0.019	92%	0.136	46%	0.112	56%	0.116	54%	0.112	56%
15	0.339	0.332	2%	0.055	84%	0.289	15%	0.239	30%	0.272	20%	0.261	23%
16	0.547	0.663	21%	0.188	66%	0.534	2%	0.485	11%	0.564	3%	0.537	2%
17	0.621	0.829	33%	0.365	41%	0.668	8%	0.656	6%	0.719	16%	0.681	10%
18	0.711	0.894	26%	0.503	29%	0.734	3%	0.748	5%	0.781	10%	0.738	4%

Table 4: Comparison between experimental void fraction measurements and model predictions

In order to provide a more immediate comparison between the proposed models and the experimentally measured void fraction values, the corresponding results are presented in Fig. 6.

The comparison highlights that the CISE, Chisholm, and Drift Flux models provide the most accurate predictions with respect to the experimental data. The homogeneous model also shows satisfactory agreement, particularly for void fraction values lower than approximately 0.6.

Conversely, the Zivi correlation exhibits the largest deviations from the experimental measurements, resulting in the least accurate predictions among the considered models.

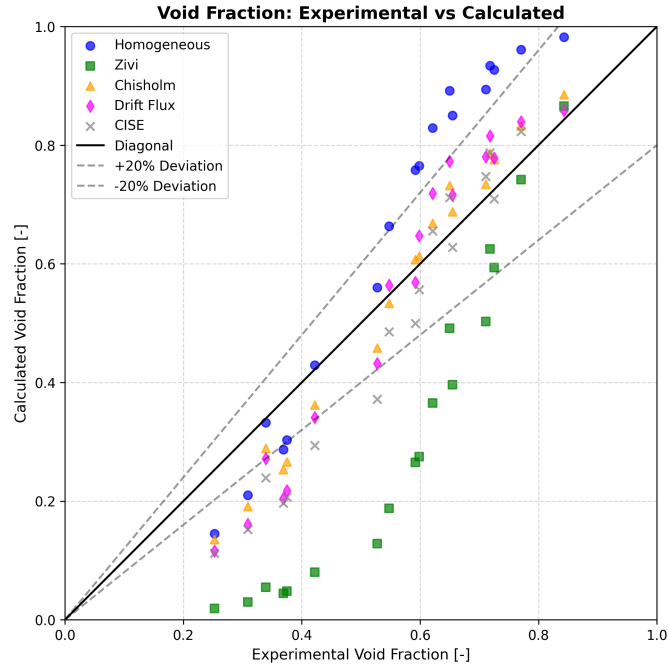


Figure 6: Void fraction comparison between experimental data and models

5 Evaluation of Experimental Pressure Difference along the Vertical Tube

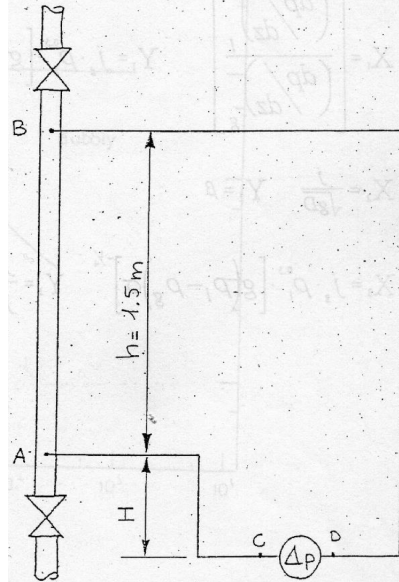


Figure 7: Scheme of the test section for pressure measurements

With reference to the evaluation of the pressure change in the test section using the differential pressure transducer, it should be noted that, when measuring the pressure variation along a vertical pipe, the pressure difference measured by the transducer does not directly correspond to the actual pressure change occurring in the test section. However, the latter can be determined by applying the principles of fluid statics and Bernoulli's equation.

The experimental setup employs a Rosemount differential pressure transducer, which is capable of measuring only positive pressure differences. Consequently, the plastic tubes connecting the pressure taps of the test section to the transducer must be arranged such that the pressure difference between the “High” and “Low” chambers of the transducer is always positive.

For this reason, different connection schemes between the pressure taps and the transducer are adopted depending on the operating conditions and flow direction. It is therefore essential to record the connection configuration used during each experimental test.

With reference to both upward and downward flows, the following sections provide guidelines for evaluating the actual pressure change in the test section as a function of the pressure difference measured between the chambers of the differential pressure transducer, for the various possible connection configurations.

With reference to Fig. 7, the pressures at points C and D can be expressed through hydrostatic considerations as:

$$p_D = p_B + \rho g(h + H) \quad (43)$$

$$p_C = p_A + \rho gH \quad (44)$$

where ρ is the density of the water contained in the plastic connecting pipes, which can be assumed equal to:

$$\rho = 1000 \text{ kg/m}^3 \quad (45)$$

Subtracting the two previous equations yields:

$$p_C - p_D = p_A + \rho gH - p_B - \rho g(h + H) = p_A - p_B - \rho gh \quad (46)$$

Hence, the actual pressure difference along the test section is given by:

$$p_A - p_B = p_C - p_D + \rho gh \quad (47)$$

In order to compare the experimental results with the theoretical models, the total pressure difference must be evaluated. Neglecting the acceleration pressure variation associated with density and void fraction changes, the total pressure difference can be written as the sum of the elevation and frictional contributions:

$$p_A - p_B = (p_A - p_B)_{\text{elev}} + (p_A - p_B)_{\text{frict}} = \bar{\rho}gh + \Delta p_{\text{frict}} \quad (48)$$

where $\bar{\rho}$ is the average density of the two-phase mixture in the test section.

Substituting Eq. (48) into Eq. (46) gives:

$$p_C - p_D = \bar{\rho}gh + \Delta p_{\text{frict}} - \rho gh = \Delta p_{\text{frict}} - (\rho - \bar{\rho})gh \quad (49)$$

From Eq. (49), the following considerations can be made:

- If:

$$\Delta p_{\text{frict}} > (\rho - \bar{\rho})gh$$

then:

$$p_C - p_D > 0$$

In this case, the lower pressure tap must be connected to the “High” chamber of the transducer, and the transducer directly measures:

$$\Delta p_{\text{transd}} = p_C - p_D$$

- If:

$$\Delta p_{\text{frict}} < (\rho - \bar{\rho})gh$$

then:

$$p_C - p_D < 0$$

In this case, the upper pressure tap must be connected to the “High” chamber, and the transducer measures:

$$\Delta p_{\text{transd}} = p_D - p_C$$

In the latter case:

$$\Delta p_{\text{transd}} = p_D - p_C = p_B - p_A + \rho gh \quad (50)$$

Substituting Eq. (48):

$$\Delta p_{\text{transd}} = -(\bar{\rho}gh + \Delta p_{\text{frict}}) + \rho gh \quad (51)$$

Therefore:

$$\bar{\rho}gh + \Delta p_{\text{frict}} = \rho gh - \Delta p_{\text{transd}} \quad (52)$$

Taking into account all the previous considerations, the following correlations must be used for the evaluation of the pressure change in the test section:

- If:

$$\Delta p_{\text{transd}} = p_C - p_D$$

then:

$$p_A - p_B = \bar{\rho}gh + \Delta p_{\text{frict}} = \Delta p_{\text{transd}} + \rho gh \quad (53)$$

- If:

$$\Delta p_{\text{transd}} = p_D - p_C$$

then:

$$p_A - p_B = \bar{\rho}gh + \Delta p_{\text{frict}} = \rho gh - \Delta p_{\text{transd}} \quad (54)$$

The following results are botained:

Test #	Type of differential pressure	Pressure drop (Pa)
1	pC - pD	-
2	pD - pC	1,02E+04
3	pD - pC	8,22E+03
4	pD - pC	7,45E+03
5	pD - pC	9,15E+03
6	pD - pC	5,08E+03
7	pD - pC	5,43E+03
8	pD - pC	5,70E+03
9	pD - pC	1,20E+04
10	pD - pC	1,09E+04
11	pD - pC	9,57E+03
12	pD - pC	1,17E+04
13	pD - pC	-
14	pD - pC	1,44E+04
15	pD - pC	1,38E+04
16	pD - pC	1,37E+04
17	pC - pD	1,69E+04
18	pC - pD	2,11E+04

Table 5: Experimental pressure difference along the vertical pipe, using the difference of pressure measured by a differential pressure transducer

5.1 Elevation Pressure Drop with Void Fraction α

The pressure drop in a two-phase flow can be evaluated by applying the momentum conservation equation. Accordingly, the following expression is obtained:

$$-\frac{dP}{dz} = \frac{\tau_w P}{A} + \rho_h g \sin \theta + G^2 \frac{d}{dz} \left(\frac{1}{\rho_h} \right) \quad (55)$$

The first term is referred to friction, the second is the static head and the third one is related to acceleration [3]. In particular, when a separated two-phase flow model is considered, as in the case of air–water flow, the momentum equation can be expressed as:

$$-\frac{dp}{dz} = - \left(\frac{dp}{dz} \right)_F + g \sin \theta [(1 - \alpha)\rho_l + \alpha\rho_g] + G^2 \frac{d}{dz} \left[\frac{(1 - x)^2}{\rho_l(1 - \alpha)} + \frac{x^2}{\rho_g\alpha} \right] \quad (56)$$

Assuming negligible frictional effects (due to plexiglass) and fully developed steady-state flow conditions, the frictional and accelerational terms can be neglected. In particular, this implies negligible wall and interfacial shear stresses, as well as constant flow quality and void fraction along the axial direction. Consequently, the momentum equation reduces to the gravitational contribution only:

$$-\frac{dp}{dz} = g \sin \theta [(1 - \alpha)\rho_l + \alpha\rho_g] \quad (57)$$

By using the void fraction values previously calculated through the models described in Section 3, the following elevation pressure drops are obtained, depending on the adopted void fraction correlation.

Test #	Homogeneous	Zivi	Chisholm	CISE	Drift flux (bubbly)	Drift flux (plug)
1	1,03E+04	1,40E+04	1,08E+04	1,17E+04	1,15E+04	1,15E+04
2	1,05E+04	1,41E+04	1,10E+04	1,18E+04	1,17E+04	1,17E+04
3	6,48E+03	1,28E+04	7,98E+03	9,25E+03	8,37E+03	8,52E+03
4	2,22E+03	8,89E+03	4,61E+03	5,49E+03	4,19E+03	4,67E+03
5	5,92E+02	3,81E+03	2,48E+03	2,62E+03	2,38E+03	3,06E+03
6	2,80E+02	2,00E+03	1,71E+03	2,03E+03	2,10E+03	2,80E+03
7	1,10E+03	5,99E+03	3,32E+03	4,29E+03	3,28E+03	3,79E+03
8	3,58E+03	1,08E+04	5,79E+03	7,37E+03	6,36E+03	6,50E+03
9	1,16E+04	1,43E+04	1,19E+04	1,25E+04	1,24E+04	1,24E+04
10	8,41E+03	1,35E+04	9,39E+03	1,04E+04	9,70E+03	9,86E+03
11	3,47E+03	1,07E+04	5,71E+03	6,54E+03	5,21E+03	5,65E+03
12	1,61E+03	7,49E+03	3,96E+03	4,26E+03	3,36E+03	3,96E+03
13	9,92E+02	5,53E+03	3,16E+03	3,14E+03	2,73E+03	3,39E+03
14	1,26E+04	1,44E+04	1,27E+04	1,31E+04	1,30E+04	1,31E+04
15	9,84E+03	1,39E+04	1,05E+04	1,12E+04	1,07E+04	1,09E+04
16	4,97E+03	1,20E+04	6,87E+03	7,59E+03	6,43E+03	6,83E+03
17	2,53E+03	9,35E+03	4,90E+03	5,09E+03	4,16E+03	4,72E+03
18	1,59E+03	7,33E+03	3,93E+03	3,73E+03	3,25E+03	3,88E+03

Table 6: Elevation pressure changes obtained using different void fraction models

The figure compares pressure drops predicted by five theoretical models with experimental measurements. The solid black diagonal represents the ideal agreement, while the dashed grey lines define a $\pm 20\%$ error band.

Overall, the Drift Flux, Drift Bubbly, Chisholm, and CISE models show strong predictive performance, with most data points lying within the acceptable error range and closely following the reference line. In contrast, the Homogeneous model (blue markers) systematically underpredicts the experimental values, with deviations increasing as the pressure drop rises and frequently exceeding the lower -20% .

5.2 Frictional Pressure Drop with Friedel Correlation

The frictional pressure drop is no longer neglected. To determinate this quantity several models were developed, based on theory but integrated by parameters determined experimentally. The Friedel correlation for vertical upward two-phase flow is expressed as:

$$\phi_{l0}^2 = E + \frac{3.24 F H}{f_{g0}^{0.045} Fr_h^{0.045} We_h^{0.035}} \quad (58)$$

where:

$$E = (1 - x)^2 + x^2 \frac{\rho_l f_{g0}}{\rho_g f_{l0}} \quad (59)$$

The parameters f_{g0} and f_{l0} are the friction factors calculated by considering the total mass flow rate flowing entirely as gas or entirely as liquid, respectively.

The correction factor F is defined as:

$$F = x^{0.78} (1 - x)^{0.224} \quad (60)$$

The parameter H is given by:

$$H = \left(\frac{\rho_l}{\rho_g} \right)^{0.91} \left(\frac{\mu_g}{\mu_l} \right)^{0.19} \left(1 - \frac{\mu_g}{\mu_l} \right)^{0.7} \quad (61)$$

The homogeneous Froude number is defined as:

$$Fr_h = \frac{G^2}{gD\rho_h^2} \quad (62)$$

The homogeneous Weber number is defined as:

$$We_h = \frac{G^2 D}{\sigma \rho_h} \quad (63)$$

Finally, the homogeneous density is evaluated as:

$$\rho_h = \left(\frac{x}{\rho_g} + \frac{1 - x}{\rho_l} \right)^{-1} \quad (64)$$

The following results are obtained:

f_{go}	f_{lo}	E	F	H	Fr_h	We_h	Multiplier	$-(dp/dz)$	ΔP (Pa)
6.40E-02	2.77E-02	9.99E-01	3.40E-03	1.66E+02	2.96E+00	1.90E+02	2.45E+00	4.78E+02	7.18E+02
6.53E-02	2.78E-02	9.99E-01	3.19E-03	1.66E+02	2.80E+00	1.84E+02	2.36E+00	4.60E+02	6.90E+02
4.91E-02	2.81E-02	1.00E+00	7.75E-03	1.67E+02	7.40E+00	3.00E+02	4.14E+00	8.18E+02	1.23E+03
3.38E-02	2.83E-02	1.04E+00	2.47E-02	1.66E+02	6.36E+01	8.86E+02	9.72E+00	1.96E+03	2.94E+03
2.34E-02	2.86E-02	1.66E+00	7.62E-02	1.64E+02	9.41E+02	3.49E+03	2.40E+01	5.12E+03	7.69E+03
2.33E-02	3.46E-02	3.59E+00	1.35E-01	1.68E+02	9.58E+02	1.68E+03	4.52E+01	2.66E+03	3.99E+03
3.38E-02	3.45E-02	1.19E+00	4.52E-02	1.68E+02	5.52E+01	3.79E+02	1.79E+01	9.27E+02	1.39E+03
4.80E-02	3.45E-02	1.01E+00	1.54E-02	1.68E+02	5.03E+00	1.13E+02	7.63E+00	3.83E+02	5.75E+02
6.34E-02	2.53E-02	9.99E-01	2.33E-03	1.60E+02	6.57E+00	4.78E+02	1.90E+00	9.63E+02	1.44E+03
4.89E-02	2.53E-02	9.99E-01	5.20E-03	1.61E+02	1.26E+01	6.64E+02	2.93E+00	1.50E+03	2.25E+03
3.39E-02	2.53E-02	1.01E+00	1.63E-02	1.61E+02	7.46E+01	1.62E+03	6.42E+00	3.30E+03	4.96E+03
2.68E-02	2.54E-02	1.09E+00	3.38E-02	1.59E+02	3.49E+02	3.52E+03	1.11E+01	5.78E+03	8.66E+03
2.33E-02	2.54E-02	1.24E+00	5.18E-02	1.56E+02	9.22E+02	5.72E+03	1.54E+01	8.07E+03	1.21E+04
6.34E-02	2.23E-02	1.00E+00	1.59E-03	1.66E+02	1.49E+01	1.17E+03	1.59E+00	1.90E+03	2.85E+03
4.81E-02	2.24E-02	9.99E-01	3.75E-03	1.62E+02	2.44E+01	1.50E+03	2.32E+00	2.78E+03	4.17E+03
3.39E-02	2.24E-02	1.00E+00	1.13E-02	1.57E+02	9.45E+01	2.94E+03	4.54E+00	5.38E+03	8.07E+03
2.69E-02	2.25E-02	1.03E+00	2.32E-02	1.53E+02	3.60E+02	5.70E+03	7.55E+00	8.86E+03	1.33E+04
2.33E-02	2.25E-02	1.09E+00	3.65E-02	1.47E+02	9.05E+02	8.99E+03	1.04E+01	1.21E+04	1.82E+04

Table 7: Friedel multiplier and corresponding frictional pressure drops

5.3 Frictional Pressure Drop with Homogeneous Multiplier

For the homogeneous model, the two-phase frictional pressure drop is expressed through a homogeneous multiplier approach:

$$\phi_{lo}^2 = \frac{\left(\frac{dp}{dz} \right)_{TP}}{\left(\frac{dp}{dz} \right)_{lo}} = \frac{f_{TP}}{f_{lo}} \cdot \frac{\rho_h}{\rho_l} \cdot \left(1 + x \frac{\rho_l}{\rho_g} \right) \quad (65)$$

The friction factors f_{TP} and f_{lo} are evaluated at different Reynolds numbers, defined as:

$$Re_h = \frac{GD}{\mu_h}, \quad Re_{lo} = \frac{GD}{\mu_l} \quad (66)$$

Since the friction factor decreases with Reynolds number and tends to become constant in fully rough pipes at high Reynolds numbers, it is sometimes assumed that:

$$f_{TP} \approx f_{lo} \approx 1 \quad (67)$$

The results are reported in Tab. 8:

$\mu_{homogeneous}$	$Re_{homogeneous}$	f (Blasius)	-(dp/dz)	ΔP [Pa]
0,001137252	1,39E+04	2,91E-02	2,95E+02	4,43E+02
0,001153375	1,36E+04	2,93E-02	2,87E+02	4,31E+02
0,001039983	1,51E+04	2,85E-02	4,56E+02	6,84E+02
0,000487686	3,25E+04	2,35E-02	1,11E+03	1,67E+03
0,000154648	1,05E+05	1,76E-02	3,26E+03	4,89E+03
7,96825E-05	9,71E+04	1,79E-02	1,60E+03	2,40E+03
0,000261817	2,78E+04	2,45E-02	4,94E+02	7,41E+02
0,000721521	9,92E+03	3,17E-02	1,90E+02	2,85E+02
0,001311358	2,03E+04	2,65E-02	6,74E+02	1,01E+03
0,001295339	2,06E+04	2,64E-02	9,33E+02	1,40E+03
0,00076105	3,51E+04	2,31E-02	1,99E+03	2,98E+03
0,000400135	6,70E+04	1,96E-02	3,68E+03	5,52E+03
0,000259133	1,04E+05	1,76E-02	5,37E+03	8,05E+03
0,001260378	3,44E+04	2,32E-02	1,45E+03	2,18E+03
0,001339489	3,24E+04	2,36E-02	1,88E+03	2,83E+03
0,000990314	4,35E+04	2,19E-02	3,43E+03	5,14E+03
0,000585867	7,32E+04	1,92E-02	5,83E+03	8,75E+03
0,000388965	1,09E+05	1,74E-02	8,31E+03	1,25E+04

Table 8: Homogeneous multiplier and corresponding frictional pressure drops

5.4 Total Pressure Drop

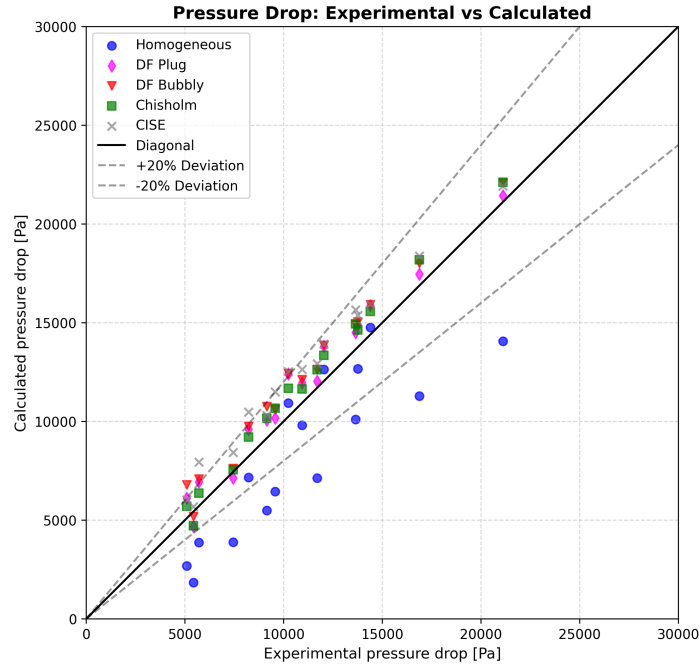


Figure 8: Pressure drop comparison between experimental data and models

Figure 8 reports the comparison between the experimental pressure drops and the values predicted by the different model combinations. The results confirm that the total pressure-drop estimate is strongly affected by the combined evaluation of the gravitational and frictional contributions. In particular, the homogeneous model systematically underestimates the pressure drop, especially as the gas flow rate increases and the flow regime moves toward more complex slug, churn, or annular configurations. This behavior is consistent with the simplified assumptions of the homogeneous approach, which do not fully account for phase slip and for the actual distribution of the phases inside the pipe.

The predictions based on the Friedel frictional correlation show a clear improvement with respect to the homogeneous multiplier, since they provide a more realistic description of the frictional amplification caused by two-phase flow. The best agreement is obtained when the Friedel model is coupled with void-fraction correlations that already showed strong accuracy in the previous analysis. In particular, the Chisholm correlation and the Drift Flux formulation confirm their reliability, because their treatment of phase slip leads to more consistent elevation contributions and, consequently, to more accurate total pressure-drop predictions.

Conclusions

The present experimental investigation addressed the characterization of upward air–water two-phase flow in a vertical pipe, with particular emphasis on flow pattern identification, void fraction evaluation, and pressure drop prediction. The study combined direct experimental measurements with a systematic comparison against a wide range of theoretical and semi-empirical models.

From a qualitative standpoint the observed flow regimes, ranging from bubbly and slug flow to churn and annular configurations highlighted the strong dependence of flow structure on the imposed gas and liquid superficial velocities. The comparison with classical flow pattern maps, such as Hewitt–Roberts and Taitel Duckler, showed an overall reasonable agreement in terms of dominant regimes, although some discrepancies were observed. These deviations underline the intrinsic limitations of flow-pattern maps, particularly in the transition regions where visual classification remains partially subjective and sensitive to experimental interpretation.

The analysis of void fraction revealed a clear hierarchy in model performance. The CISE, Chisholm, and Drift Flux formulations provided the most accurate predictions across the widest range of operating conditions, confirming their robustness in accounting for phase slip effects. The homogeneous model, while computationally simpler, showed acceptable agreement only at relatively low void fractions, where velocity differences between phases are limited. Conversely, the Zivi correlation systematically underestimated the experimental data, indicating that its applicability is mainly limited to high-void-fraction conditions.

The pressure drop analysis further emphasized the importance of evaluating the gravitational and frictional contributions together, rather than treating elevation effects as a stand-alone prediction method. A more structured comparison of the combined models showed that the accuracy of the total pressure-drop estimate depends strongly on the coupling between the void-fraction correlation and the frictional multiplier. In this respect, the combinations based on the Friedel frictional correlation provided the most consistent agreement with the experimental data, especially when coupled with the Chisholm void-fraction model or with the Drift Flux plug-flow formulation.

Overall, the results demonstrate that no single model is universally valid across all flow regimes and operating conditions. Instead, predictive accuracy depends strongly on the underlying physical assumptions of each formulation, particularly regarding phase distribution, slip ratio, interfacial interactions, and frictional amplification. Among the tested pressure-drop combinations, Friedel-Chisholm and Friedel-Drift Flux plug emerged as the most accurate and robust options for the investigated range, while the homogeneous based combinations showed larger deviations, particularly when gas flow rates increased and flow-regime transitions became more pronounced.

Finally, the study confirms the importance of integrated experimental and theoretical analysis in two-phase flow systems. While empirical correlations remain essential tools for engineering applications, their limitations must be carefully considered, especially in complex transitional regimes.

AI Acknowledgment

Artificial Intelligence tools, specifically Gemini Pro and Prism, were used as auxiliary support during the preparation of this report. Their use was limited to assisting the authors in improving the quality of graphical contributions and to proofreading the written text. All technical analyses, model applications, numerical evaluations, interpretation of results, and final conclusions remain the responsibility of the authors.

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